

# Entropy's Blind Spot: Patterns



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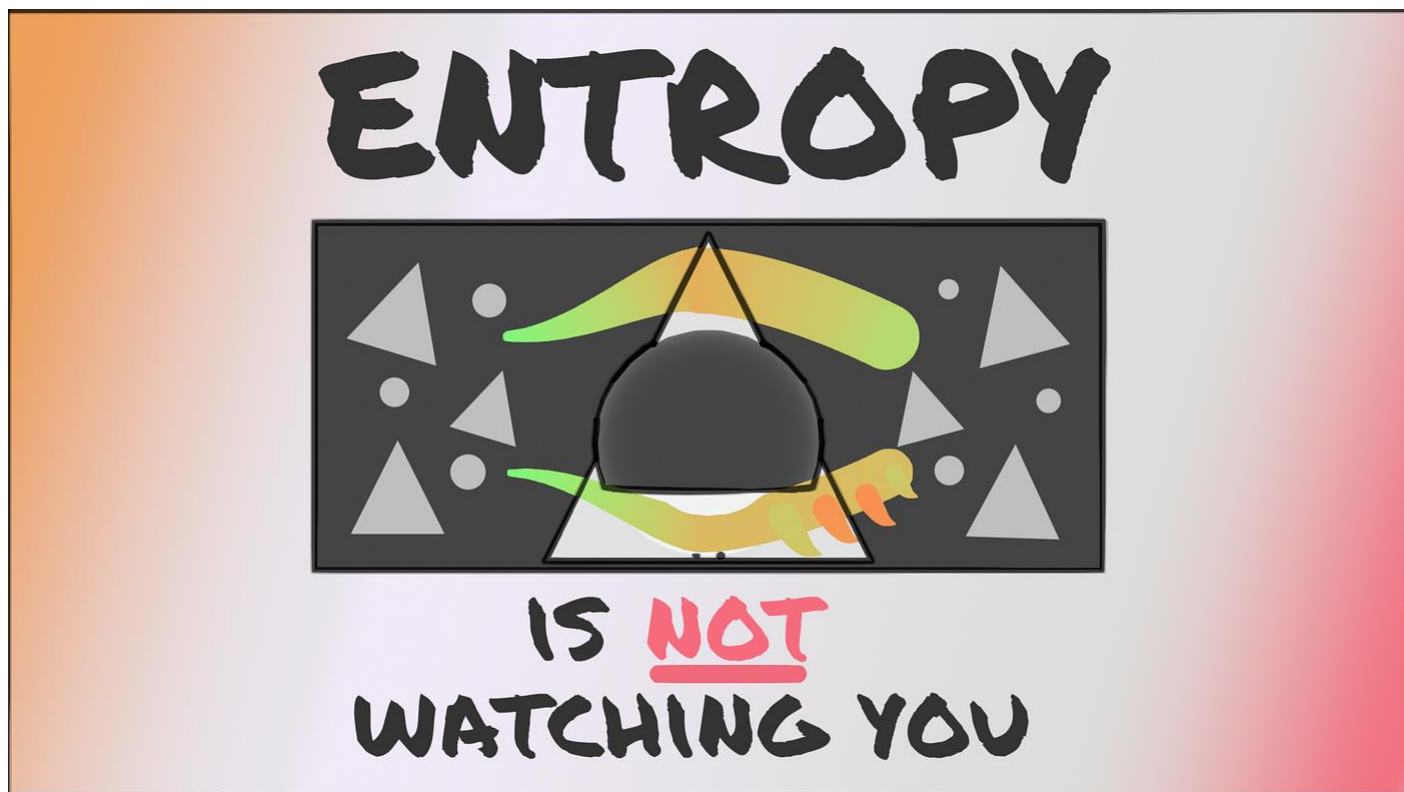
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Shannon's entropy aims to measure the number of bits needed to represent a set of microstates, but it fails when there are internal patterns, order, or a "Maxwell demon".

It presumes that each symbol is chosen at random and that there are no correlations between successive symbols.

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**But correlations are precisely where patterns live.** Because entropy ignores these dependencies, its “blind spot” means it can miss the very regularities that reduce the actual number of bits needed to describe the microstates.

The limits

The goal of entropy is to measure the **smallest number of bits needed to store data without losing information.**

It works well for individual symbol study, when there are no internal patterns in the microstates; it assumes the sequences are fairly homogeneous.

The **primary issue** is that **entropy fails when patterns are present.**

The blind formula

To understand why this happens, we can check the formula of entropy:  
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$$H(P_i) = \sum_{i=1}^n P_i \log_2(P_i)$$

2  
→ bits

To see this, we can compare the entropy of sequences

like 000000111111, 001100110011, 010101010101, 110100111000. **Their entropy is the same**, but intuition suggests that, for instance, 000000111111 looks less chaotic (or contains less information) than 110100111000. Let's explore why this happens.

In the formula, we can see it only cares about the frequency of each symbol alone, not in pairs of **n-grams (four 1 together)**, which form the patterns. This is why, as long as we have the same number of repetitions by symbols, it will result in the same entropy amount.

Swaps invariance

So, the entropy does not care about the internal order, but only cares about the frequency of each symbol in **1-grams**. This means that **we can swap positions and the entropy will remain the same**.

This invariance is possible thanks to that each symbol is independent of the other, and there is no correlation between successive symbols, at least from the assumptions of the entropy definition; **if an internal correlation exists, then entropy will be blind to it**.

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The same entropy calculation from different permutations

At the end, the **entropy is the same** for all the permutations that have the same instances of the symbols (same frequency). We can group them, given the amount of 1, for example, in a sequence of 8 symbols, all the permutations that have 3 times 1 and 5 times 0 will have the same entropy, no matter the order.

In physical systems, this also applies; the entropy is designed for steady states. If we had patterns, internal order, or a “Maxwell demon”, the entropy definition fails to count it.

**Note to remember:** To clarify how entropy is related to **chaos**, what entropy measures is the number of different configurations a system can have after random swaps. Not how chaotic a single permutation `00000111` can be, but the number of permutations (swaps) of their symbols’ order.

**Next related story:** Compression (Huffman Coding, n-gram concept), the essence of Markov chains, or recursive sequences (correlation between successive symbols).

Thanks for reading!

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